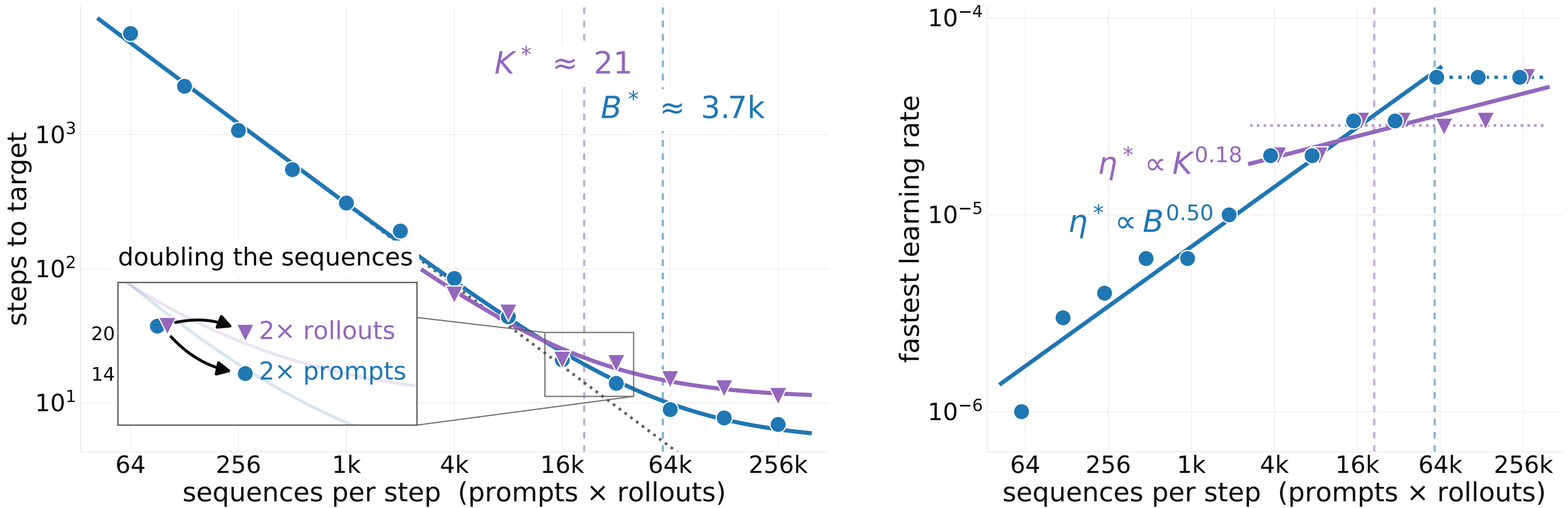
$K^* \approx 21$
$B^* \approx 3.7\text{k}$
steps to target
10^3
10^2
10^1
doubling the sequences
20
14
2× rollouts
2× prompts
64
256
1k
4k
16k
64k
256k
sequences per step (prompts × rollouts)
fastest learning rate
10^{-4}
10^{-5}
10^{-6}
$\eta^* \propto K^{0.18}$
$\eta^* \propto B^{0.50}$
64
256
1k
4k
16k
64k
256k
sequences per step (prompts × rollouts)
scaling prompts (K = 16)
scaling rollouts (B = 1024)
fit $S = S_{\min}(1 + N^*/N)$
critical batch N^* (B^* or K^*)
perfect scaling (steps $\propto 1/N$)